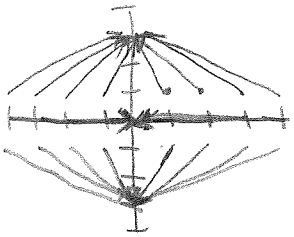
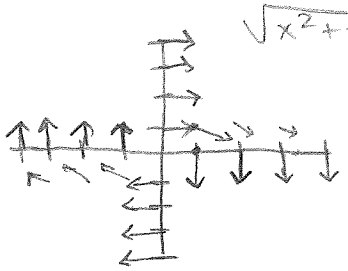
HW7

17.1 p. 1080 #4, 6, 8, 21*, 23, 26

4. $\vec{F}(x,y) = -x\uparrow + 2y\uparrow$



6. $\vec{F}(x,y) = \dfrac{y\uparrow - x\uparrow}{\sqrt{x^2+y^2}}$



all unit vectors

tangent to circles

8. $\vec{F}(x,y,z) = z\uparrow$

length z
parallel to y-axis
> xy plane, pt to +y
< xy plane, pt to -y

21. $f(x,y) = \ln(x+2y)$

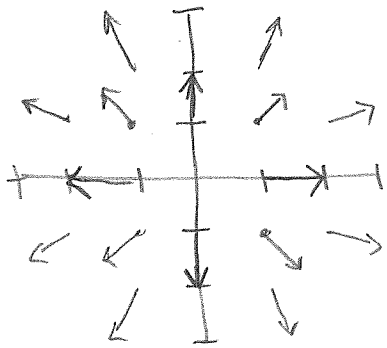
$\nabla f = \left\langle \dfrac{1}{x+2y}, \dfrac{2}{x+2y} \right\rangle$

23. $f(x,y,z) = \sqrt{x^2+y^2+z^2}$

$\nabla f = \left\langle \dfrac{x}{\sqrt{x^2+y^2+z^2}}, \dfrac{y}{\sqrt{x^2+y^2+z^2}}, \dfrac{z}{\sqrt{x^2+y^2+z^2}} \right\rangle$

26. $f(x,y) = \ln\sqrt{x^2+y^2}$

$\nabla f = \left\langle \dfrac{x}{x^2+y^2}, \dfrac{y}{x^2+y^2} \right\rangle$



HW1

17.2 p.1091 # 1, 3, 7, 13, 17, 19, 21

1. $\int_C y\,ds$, C: $x=t^2$, $y=t$, $0\le t\le 2$

$x'=2t$, $y'=1$

$$\int_0^2 t\sqrt{4t^2+1}\,dt = \frac{1}{8}\int_1^{17} u^{1/2}\,du = \frac{1}{8}\cdot\frac{2}{3}\left(17^{3/2}-1\right)$$

$$= \frac{1}{12}\left(17^{3/2}-1\right)$$

3. $\int_C xy^4\,ds$, right half of $x^2+y^2=16$

$x=4\cos t$ $y=4\sin t$ $-\frac{\pi}{2}\le t\le\frac{\pi}{2}$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 4\cos t\,(4\sin t)^4\sqrt{16\cos^2 t+16\sin^2 t}\,dt$$

$$4^6\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos t\cdot\sin^4 t\,dt = 4^6\int_{-1}^{1} u^4\,du = 4^6\cdot\frac{1}{5}\left(1^5-(-1)^5\right)$$

$$= 4^6\cdot\frac{2}{5} = 1638.4$$

7. $\int_C xy\,dx + (x-y)\,dy$ (0,0) to (2,0) to (3,2)

C_1: $x=2t$, $y=0$ $0\le t\le 1$

C_2: $x=2+t$, $y=2t$

$$\int_0^1 \left(2t(0)(2)+(2t-0)(0)\right)dt + \int_0^1 \left((2+t)(2t)(1)+(2+t-2t)(2)\right)dt$$

$$= \int_0^1 \left[4t+2t^2+4-2t\right]dt = \int_0^1 \left(2t^2+2t+4\right)dt$$

$$= \left[\frac{2}{3}t^3+t^2+4t\right]\Big|_0^1 = \frac{2}{3}+1+4 = 17/3$$

13. $\int_C x^3y^2z\,dz$ C: $x=2t$, $y=t^2$, $z=t^2$, $0\le t\le 1$

$$\int_0^1 \left[(2t)^3(t^2)^2(t^2)\cdot 2t\,dt + 0\,dx + 0\,dy\right]$$

$$\int_0^1 8t^3\cdot t^4\cdot t^2\cdot 2t\,dt = 16\int_0^1 t^{10}\,dt = \frac{16}{11}$$

17. a) $(-3,-3)$ to $(-3,3)$
positive, mostly same direction

b) circle $r=3$
negative

19. $\vec{F} = x^2y^3\uparrow - y\sqrt{x}\,\uparrow$ $\quad \vec{r} = t^2\uparrow - t^3\uparrow$ $\quad 0\le t\le 1$

$r' = 2t\uparrow - 3t^2\uparrow$

$\vec{F}(\vec{r}) = t^4\cdot(-t^3)^3\uparrow + t^3\sqrt{t^2}\,\uparrow$

$= -t^{13}\uparrow + t^4\uparrow$

$$\int_0^1 (-2t^{14} - 3t^6)\,dt = \left(-\frac{2}{15} - \frac{3}{7}\right) = -59/105$$

21. $\vec{F} = \sin x\uparrow + \cos y\uparrow + xz\hat{k}$, $\vec{r} = t^3\uparrow - t^2\uparrow + t\hat{k}$ $\quad 0\le t\le 1$

$$\int_0^1 (\sin t^3\cdot 3t^2 + \cos(-t^2)(-2t) + t^4\cdot 1)\,dt$$

$$\int_0^1 \sin u\,du + \int_0^{-1} \cos u\,du + \int_0^1 t^4\,dt$$

$$-\cos u\Big|_0^1 + \sin u\Big|_0^{-1} + \frac{1}{5} = -\cos 1 + \cos 0 + \sin(-1) - \sin 0 + \frac{1}{5}$$

$$= -(\cos 1 + \sin 1) + \frac{6}{5}$$

HW7

17.3 p.1100 #5, 7, 13, 17, 19, 21, 23, 28

5. $\vec{F} = xe^y\uparrow + ye^x\uparrow$

$\frac{\partial P}{\partial y} = xe^y \qquad \frac{\partial Q}{\partial x} = ye^x \qquad \therefore$ not conservative

7. $\vec{F} = (2x\cos y - y\cos x)\uparrow + (-x^2\sin y - \sin x)\uparrow$

$\frac{\partial P}{\partial y} = -2x\sin y - \cos x \qquad \frac{\partial Q}{\partial x} = -2x\sin y - \cos x \therefore$ yes, cons.

$\int (2x\cos y - y\cos x)\,dx = x^2\cos y - y\sin x + g(y) + C$
$\int (-x^2\sin y - \sin x)\,dy = x^2\cos y - y\sin x + h(x) + C$

$f(x,y) = x^2\cos y - y\sin x + C$

13. $\vec{F} = x^3y^4\uparrow + x^4y^3\uparrow$ C: $\vec{r}(t) = \sqrt{t}\uparrow + (1+t^3)\uparrow,\ 0\le t\le 1$

$\int x^3y^4\,dx = \frac{1}{4}x^4y^4 + \vec{C}$

$\int x^4y^3\,dy = \frac{1}{4}x^4y^4 + \vec{C}$

a) $f(x,y) = \frac{1}{4}x^4y^4 + C$

$\therefore$ conservative and $\int_C F\cdot dr = f\Big|_a^b$

$\vec{r}(0) = \langle 0,1\rangle$
$\vec{r}(1) = \langle 1,2\rangle$

b) $\int_C \vec{F}\cdot d\vec{r} = f(1,2) - f(0,1)$
$= \frac{1}{4}\cdot 1\cdot 16 - \frac{1}{4}\cdot 0\cdot 1 = 4$

17. $\vec{F} = (2xz + \sin y)\uparrow + x\cos y\uparrow + x^2\hat{k}$ C: $\vec{r} = \langle \cos t, \sin t, t\rangle$
$0\le t\le 2\pi$

$\int (2xz + \sin y)\,dx = x^2z + x\sin y$
$\int (x\cos y)\,dy = x\sin y$
$\int x^2\,dz = x^2z$

a) $f(x,y,z) = x^2z + x\sin y + C$

$\vec{r}(0) = \langle 1,0,0\rangle$
$\vec{r}(2\pi) = \langle 1,0,2\pi\rangle$

b) $f(1,0,2\pi) - f(1,0,0)$
$= (2\pi + 0) - (0 + 0) = 2\pi$

19. $\int_C (2x\sin y\,dx + (x^2\cos y - 3y^2)dy)$ $(-1,0)$ to $(5,1)$

$\frac{\partial P}{\partial y} = 2x\cos y$ $\frac{\partial Q}{\partial x} = 2x\cos y$ $\therefore$ cons. $\therefore$ indep.

$\int (2x\sin y)dx = x^2\sin y$ $f(x,y) = x^2\sin y - y^3$
$\int (x^2\cos y - 3y^2)dy = x^2\sin y - y^3$

$f(5,1) - f(-1,0) = (25\sin 1 - 1) - (\sin 0 - 0)$
$= 25\sin(1) - 1$

21. $\vec{F} = x^2y^3\uparrow + x^3y^2\uparrow$ $(0,0)$ to $(2,1)$

$\int x^2y^3\,dx = \frac{1}{3}x^3y^3$ $f(x,y) = \frac{1}{3}x^3y^3$
$\int x^3y^2\,dy = \frac{1}{3}x^3y^3$ $f(2,1) - f(0,0) = \frac{8}{3} - 0 = \frac{8}{3}$

23. if conservative, $\int_C \vec{F}\cdot d\vec{r} = 0$ on a closed C
but circle w/ center as origin going counter clockwise would be negative
So No

28. $\int ydx + xdy + xyzdz$

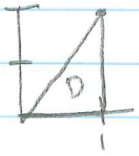
$\frac{\partial P}{\partial y} = 1$ $\frac{\partial Q}{\partial x} = 1$ $\therefore$ not cons.
$\therefore$ not indep.

$\frac{\partial P}{\partial z} = 0$ $\frac{\partial R}{\partial x} = yz$ $\rightarrow$ not =

$\frac{\partial Q}{\partial z} = 0$ $\frac{\partial R}{\partial y} = xz$ $\rightarrow$ not =

HW 7

17.4 p. 1108 # 3, 7*, 11, 15, 17, 19



3. $\oint_C (xy\,dx + x^2y^3\,dy)$ C is △ w/ (0,0) (1,0) (1,2)

C_1: $x=t$, $y=0$ $\quad 0 \le t \le 1$

C_2: $x=1$, $y=t$ $\quad 0 \le t \le 2$

C_3: $x=t$, $y=2t$ $\quad 1 \ge t \ge 0$

$$\int_0^1 (0+0)\,dt + \int_0^2 (t\cdot 0 + t^3)\,dt + \int_1^0 (2t^2 + 8t^5\cdot 2)\,dt$$

$$= \int_0^2 t^3\,dt - \int_0^1 (2t^2 + 16t^5)\,dt$$

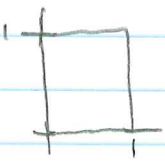
$$= \frac{1}{4}(16) - \left(\frac{2}{3} + \frac{16}{6}\right) = 4 - \frac{10}{3} = \frac{2}{3}$$

$$\iint_D (2xy^3 - x)\,dA = \int_0^1 \int_0^{2x} (2xy^3 - x)\,dy\,dx$$

$$= \int_0^1 \left(\frac{1}{2}xy^4 - xy\right)\Big|_0^{2x} dx$$

$$= \int_0^1 (8x^5 - 2x^2)\,dx$$

$$= \frac{8}{6} - \frac{2}{3} = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$



7. $\oint_C (e^y\,dx + 2xe^y\,dy)$ square: $x=0, x=1, y=0, y=1$

$$= \iint_D (2e^y - e^y)\,dA = \int_0^1 \int_0^1 e^y\,dx\,dy = \int_0^1 e^y\,dy = e^y\Big|_0^1$$

$$= e - 1$$

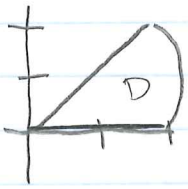


11. $\oint_C (y^3\,dx - x^3\,dy)$ circle: $x^2 + y^2 = 4$

$$= \iint_D (-3x^2 - 3y^2)\,dA = -3\int_0^{2\pi} \int_0^2 (r^2\cos^2\theta + r^2\sin^2\theta)\,r\,dr\,d\theta$$

$$= -3\int_0^{2\pi} d\theta \int_0^2 r^3\,dr = -3(2\pi)\left(\frac{1}{4}\cdot 2^4\right) = -6\pi(4)$$

$$= -24\pi$$



15. $\oint_C \vec{F}\cdot d\vec{r}$ $\quad \vec{F} = (y^2 - x^2y)\uparrow + xy^2\uparrow$

$= \iint_D (y^2 - (2y - x^2))\,dA = \iint_D (x^2 + y^2 - 2y)\,dA$

$= \int_0^{\pi/4}\int_0^2 (r^2 - 2r\sin\theta)\, r\, dr\, d\theta$

$= \int_0^{\pi/4}\int_0^2 (r^3 - 2r^2\sin\theta)\, dr\, d\theta$

$= \int_0^{\pi/4} \left(\frac{1}{4}r^4 - \frac{2}{3}r^3\sin\theta\right)\Big|_0^2 d\theta$

$= \int_0^{\pi/4} \left(4 - \frac{16}{3}\sin\theta\right) d\theta = \left(4\theta + \frac{16}{3}\cos\theta\right)\Big|_0^{\pi/4} = \pi + \frac{8\sqrt{2}}{3} - \frac{16}{3}$



17. $\vec{F} = \langle x(x+y),\ xy^2 \rangle$

$\iint_D (y^2 - x)\,dA = \int_0^1\int_0^{1-x} (y^2 - x)\, dy\, dx$

$= \int_0^1 \left(\frac{1}{3}y^3 - xy\right)\Big|_0^{1-x} dx = \int_0^1 \left(\frac{1}{3}(1-x)^3 - x(1-x)\right) dx$

$u = 1 - x$
$du = -dx$

$= -\int_1^0 \left(\frac{1}{3}u^3 - (1-u)u\right) du = \int_0^1 \left(\frac{1}{3}u^3 - u + u^2\right) du$

$= \frac{1}{12} - \frac{1}{2} + \frac{1}{3} = \frac{1}{12} - \frac{6}{12} + \frac{4}{12} = -\frac{1}{12}$

19. $\vec{r}(t) = \cos^3 t\uparrow + \sin^3 t\uparrow, \quad 0 \le t \le 2\pi$

$A = \frac{1}{2}\oint_C (x\,dy - y\,dx) = \oint_C x\,dy = -\oint_C y\,dx$

$x = \cos^3 t$
$y = \sin^3 t$

$A = \int_0^{2\pi} \cos^3 t \cdot 3\sin^2 t \cos t\, dt$

$= 3\int_0^{2\pi} \cos^4 t \sin^2 t\, dt = \frac{3\pi}{8}$

HW8

17.5 p. 1115 # 1*, 7, 12c, d, f, l, 15, 17, 23, 30a

1. $\vec{F}(x,y,z) = xy\hat{i} + yz\hat{j} + zx\hat{k}$

a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & yz & zx \end{vmatrix} = \langle -y, -(z), -x \rangle$

curl $\vec{F} = \langle -y, -z, -x \rangle$

b) $\nabla \cdot \vec{F} = \frac{\partial}{\partial x} xy + \frac{\partial}{\partial y} yz + \frac{\partial}{\partial z} zx = y + z + x = \text{div } \vec{F}$

7. $\vec{F}(x,y,z) = \frac{x}{z}\hat{i} + \frac{y}{z}\hat{j} - \frac{1}{z}\hat{k}$

a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x/z & y/z & -1/z \end{vmatrix} = \langle 0 + \frac{y}{z^2}, -(0 + \frac{x}{z^2}), 0 - 0 \rangle$

curl $\vec{F} = \langle y/z^2, -x/z^2, 0 \rangle$

b) $\nabla \cdot \vec{F} = \frac{\partial}{\partial x}\left(\frac{x}{z}\right) + \frac{\partial}{\partial y}\left(\frac{y}{z}\right) + \frac{\partial}{\partial z}\left(\frac{-1}{z}\right) = \frac{1}{z} + \frac{1}{z} + \frac{1}{z^2}$

$\text{div } \vec{F} = \frac{2}{z} + \frac{1}{z^2} = \frac{2z+1}{z^2}$

→ 12. c) div $\vec{F}$ scalar field f) grad(div $\vec{F}$) vector field

d) curl(∇f) vector field l) div(curl(grad f)) scalar field (all 0)

curl ∇f ↳ vector

find curl? or just find f?

15. $\vec{F} = \langle 2xy, x^2 + 2yz, y^2 \rangle$

curl $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xy & x^2+2yz & y^2 \end{vmatrix} = \langle 2y - 2y, -(0-0), 2x - 2x \rangle = \vec{0}$

∴ conservative

$\int 2xy\, dx = x^2y$

$\int (x^2 + 2yz)\, dy = x^2y + y^2z$ $\quad f(x,y,z) = x^2y + y^2z + C$

$\int y^2\, dz = y^2z$

17. $\vec{F} = \langle e^x, e^z, e^y \rangle$

$$\text{curl} = \begin{vmatrix} \uparrow & \overleftarrow{\jmath} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x & e^z & e^y \end{vmatrix} = \langle e^y - e^z, -(0-0), 0-0 \rangle \neq \vec{0}$$

not conserv.

23. $\operatorname{div}(\vec{F} + \vec{G}) = \operatorname{div}\vec{F} + \operatorname{div}\vec{G}$

$\vec{F} = \langle x_1, y_1, z_1 \rangle$

$\vec{G} = \langle x_2, y_2, z_2 \rangle$

$\vec{F} + \vec{G} = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$

$$\operatorname{div}(\vec{F} + \vec{G}) = \frac{\partial}{\partial x}(x_1 + x_2) + \frac{\partial}{\partial y}(y_1 + y_2) + \frac{\partial}{\partial z}(z_1 + z_2)$$

$$= \frac{\partial}{\partial x}x_1 + \frac{\partial}{\partial x}x_2 + \frac{\partial}{\partial y}y_1 + \frac{\partial}{\partial y}y_2 + \frac{\partial}{\partial z}z_1 + \frac{\partial}{\partial z}z_2$$

$$= \operatorname{div}\vec{F} + \operatorname{div}\vec{G}$$

30a. $\vec{r} = \langle x, y, z \rangle$ $\quad r = |\vec{r}|$

$\nabla \cdot \vec{r} = ?$ $\qquad \nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$

$= 1 + 1 + 1$

$= 3$

HW8

17.6 p.1125 #3, 17, 19, 21, 27, 29, 33, 35

3. $\vec{r}(x,\theta) = \langle x, \cos\theta, \sin\theta \rangle$
cylinder w/ center x-axis, $r=1$

17. $(1,2,-3)$, $\langle 1,1,-1\rangle$, $\langle 1,-1,1\rangle$

$x = 1 + u + v$
$y = 2 + u - v$
$z = -3 - u + v$

19. $y = 6 - 3x^2 - 2z^2$, R of xz-plane
$\hookrightarrow y \geq 0$

$x = x$
$y = 6 - 3x^2 - 2z^2$ $\quad 3x^2 - 2z^2 \leq 6$
$z = z$

21. $x^2+y^2+z^2=4$, above $z=\sqrt{x^2+y^2}$ $\quad z^2+z^2=4$, $z^2=2$, or $x^2+y^2=2$

$x = x$ $\quad 2 \leq x^2+y^2 \leq 4$
$y = y$
$z = \sqrt{4-x^2-y^2}$

or $\langle 2\sin\phi\cos\theta, 2\sin\phi\sin\theta, 2\cos\phi\rangle$
$0 \leq \phi \leq \frac{\pi}{4}$, $0 \leq \theta \leq 2\pi$

27. a) reverse spiral
b) tighter spiral

29. $x = u+v$, $y = 3u^2$, $z = u-v$ $\quad (2,3,0)$

$\vec{r}_u = \langle 1, 6u, 1\rangle$

$\vec{r}_v = \langle 1, 0, -1\rangle$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \uparrow & \jmath & \hat{k} \\ 1 & 6u & 1 \\ 1 & 0 & -1 \end{vmatrix} = \langle -6u, 2, -6u\rangle$$

$u = v = 1$ $\qquad \langle -6, 2, -6\rangle$

$$-6(x-2) + 2(y-3) - 6(z-0) = 0$$
$$-6x + 2y - 6z + 6 = 0$$
$$3x - y + 3z = 3$$

33. $x + 2y + z = 4$, inside $x^2 + y^2 = 4$

$z = 4 - x - 2y$ $\qquad \iint_D \sqrt{1 + (-1)^2 + (-2)^2}\, dA = \iint_D \sqrt{6}\, dA$

D: circle w/ $x^2 + y^2 \le 4$ $\qquad = \sqrt{6} \iint_D dA$

$$= \sqrt{6}\,(\pi 2^2) = 4\pi\sqrt{6}$$

35. $z = y^2 - x^2$, between $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$

$$\iint_D \sqrt{1 + (-2x)^2 + (2y)^2}\, dA = \iint_D \sqrt{1 + 4(x^2+y^2)}\, dA$$

$$\int_0^{2\pi}\int_1^2 r\sqrt{1+4r^2}\, dr\, d\theta = 2\pi \int_5^{17} \frac{1}{8} u^{1/2}\, du$$

$$= \frac{\pi}{4} \cdot \frac{2}{3} u^{3/2} \Big|_5^{17}$$

$$= \frac{\pi}{6}\left(17^{3/2} - 5^{3/2}\right)$$

HW 8

17.7 p. 1137 # 5*, 11, 13, 15, 19, 21*, 25

5. $\iint_S x^2yz\,dS$, $z = 1+2x+3y$ $[0,3]\times[0,2]$

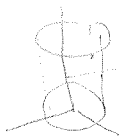
$\int_0^3 dx \int_0^2 dy \cdot x^2y(1+2x+3y)\sqrt{1+4+9}$

$\sqrt{14}\int_0^3\int_0^2 (x^2y + 2x^3 + 3x^2y^2)\,dy\,dx$

$\sqrt{14}\int_0^3 (\frac{1}{2}x^2y^2 + 2x^3y + x^2y^3)\Big|_0^2\,dx = \sqrt{14}\int_0^3 (2x^2+4x^3+8x^2)\,dx$

$\sqrt{14}\left(\frac{10}{3}x^3 + x^4\right)\Big|_0^3 = \sqrt{14}(90+81) = 171\sqrt{14}$

11. $\iint_S yz\,dS$ $z = y+3$ in $x^2+y^2=1$



$\iint_S y(y+3)\sqrt{1+1^2}\,dy\,dx$

$\sqrt{2}\int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (y^2+3y)\,dx\,dy = \sqrt{2}\int_0^{2\pi}\int_0^1 (r^3\sin^2\theta + 3r^2\sin\theta)\,dr\,d\theta$

$= \sqrt{2}\int_0^{2\pi}(\frac{1}{4}\sin^2\theta + \sin\theta)\,d\theta$

$= \sqrt{2}\left(\frac{1}{8}\theta - \frac{1}{16}\sin 2\theta - \cos\theta\right)\Big|_0^{2\pi} = \sqrt{2}\left(\frac{1}{8}\right)(2\pi) = \frac{\sqrt{2}\pi}{4} = \frac{\pi}{2\sqrt{2}}$

13. $\iint_S (x^2z + y^2z)\,dS$ $x^2+y^2+z^2=4$, $z\geq 0$

$\rho = 2$ $0\leq\phi\leq\frac{\pi}{2}$

$\int_0^{\pi/2} d\phi \int_0^{2\pi} d\theta \cdot (4\sin\phi)(8\sin^2\phi\cos^2\theta\cos\phi + 8\sin^2\phi\sin^2\theta\cos\phi)$

$32\int_0^{\pi/2} d\phi \int_0^{2\pi} d\theta\ (\sin^3\phi\cos\phi)(\cos^2\theta + \sin^2\theta)$

$32\cdot 2\pi\int_0^{\pi/2}\sin^3\phi\cos\phi\,d\phi$

$32\cdot 2\pi\cdot\frac{1}{4}\sin^4\phi\Big|_0^{\pi/2} = 16\pi$

15. $\iint_S (x^2y + z^2)\,dS$ $\quad x^2+y^2=9,\ z=0,\ z=2$

$r=3$

$$3\int_0^{2\pi} d\theta \int_0^2 dz\,(3^2\sin^2\theta\cos\theta + z^2)$$

$$3\int_0^{2\pi} (18\sin^2\theta\cos\theta + \tfrac{1}{3}2^3)\,d\theta$$

$$3\left(6\sin^3\theta + \tfrac{8}{3}\theta\right)\Big|_0^{2\pi} = 16\pi$$

19. $\vec{F} = \langle xy, yz, zx\rangle$ $\quad z = 4-x^2-y^2$ $\quad 0\le x\le 1,\ 0\le y\le 1$

$$\int_0^1\int_0^1 (-xy(-2x) - yz(-2y) + zx)\,dy\,dx$$

$$\int_0^1\int_0^1 (2x^2y + 2y^2(4-x^2-y^2) + x(4-x^2-y^2))\,dy\,dx$$

$$= \int_0^1\int_0^1 (2x^2y + 8y^2 - 2x^2y^2 - 2y^4 + 4x - x^3 - xy^2)\,dy\,dx$$

$$= \int_0^1 (x^2 + \tfrac{8}{3} - \tfrac{2}{3}x^2 - \tfrac{2}{5} + 4x - x^3 - \tfrac{1}{3}x)\,dx$$

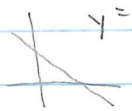
$$= \int_0^1 \left(\tfrac{1}{3}x^2 + \tfrac{11}{3}x - x^3 + \tfrac{34}{15}\right)dx = \tfrac{1}{9} + \tfrac{11}{6} - \tfrac{1}{4} + \tfrac{34}{15} = \tfrac{713}{180}$$

21. $\vec{F} = \langle xze^y, -xze^y, z\rangle$ $\quad x+y+z=1$, $x,y,z>0$

downward $\rightarrow z = 1-x-y$

$$-\iint_S (-xze^y(-1) + xze^y(-1) + z)\,dy\,dx$$

$$-\iint_S z\,dy\,dx = -\int_0^1\int_0^{1-x} (1-x-y)\,dy\,dx$$



$$-\int_0^1 \left(1-x - x(1-x) - \tfrac{1}{2}(1-x)^2\right)dx$$

$u = 1-x$, $du = -1\,dx$ $\quad x=0,\ u=1$ $\quad x=1,\ u=0$

$$-\int_0^1 \left(u - (1-u)u - \tfrac{1}{2}u^2\right)du = \int_0^1 \left(-u+u-u^2+\tfrac{1}{2}u^2\right)du$$

$$= \int_0^1 \left(-\tfrac{1}{2}u^2\right)du = -\frac{1}{6}$$

25. $\vec{F} = \langle 0, y, -z \rangle$ S_1 $y = x^2 + z^2$, $0 \le y \le 1$
S_2 $x^2 + z^2 \le 1$, $y = 1$



S_1: $\vec{r} = \langle x, x^2+z^2, z \rangle$ $\vec{r}_x = \langle 1, 2x, 0 \rangle$
$\vec{r}_z = \langle 0, 2z, 1 \rangle$

$\vec{F}(r) = \langle 0, x^2+z^2, -z \rangle$

$$\vec{r}_x \times \vec{r}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2x & 0 \\ 0 & 2z & 1 \end{vmatrix} = \langle 2x, -1, 2z \rangle$$

$$\iint_S \vec{F}(r) \cdot (\vec{r}_x \times \vec{r}_z)\, dx\, dz$$

$$\iint_S (-x^2 - z^2 - 2z^2)\, dx\, dz$$

$$-\int_0^1 r\, dr \int_0^{2\pi} d\theta \cdot (r^2 + 2r^2 \sin^2\theta)$$

$$= -\int_0^1 r^3\, dr \int_0^{2\pi} (1 + 2\sin^2\theta)\, d\theta = -\frac{1}{4}(2\pi + 2\pi)$$

$$= -\pi$$

S_2: $\vec{n} = \langle 0, 1, 0 \rangle$ $\vec{F} \cdot \vec{n} = 1$ $\vec{r} = \langle x, 1, z \rangle$
$\vec{F}(r) = \langle 0, 1, -z \rangle$

$$\iint_S 1\, dS = \text{area} = \pi(1)^2 = \pi$$

$$\iint_S \vec{F} \cdot d\vec{S} = \pi - \pi = 0$$

HW 8

17.8 p. 1143 #3, 5, 7, 9, 11a, 13, 15, 18

3. $\vec{F} = \langle x^2 e^{yz}, y^2 e^{xz}, z^2 e^{xy} \rangle$ $\quad x^2+y^2+z^2=4,\ z \geq 0$

$\iint_S \text{curl}\vec{F}\cdot d\vec{S} = \int_C \vec{F}\cdot d\vec{r}$ $\quad \vec{r} = \langle 2\cos t, 2\sin t, 0\rangle$

$\vec{r}' = \langle -2\sin t, 2\cos t, 0\rangle$

$\vec{F}(r) = \langle 4\cos^2 t + e^0, 4\sin^2 t + e^0, 0\rangle$

$= \int_0^{2\pi} (-8\cos^2 t \sin t + 8\sin^2 t \cos t)\,dt$

$= 0$

5. $\vec{F} = \langle xyz, xy, x^2yz \rangle$ $\quad$ C: square in $z=-1$

$\iint_{S_1} \text{curl}\vec{F}\cdot d\vec{S} = \oint_C \vec{F}\cdot d\vec{r} = \iint_{S_2} \text{curl}\vec{F}\cdot d\vec{S}$

S_1: box no bottom; C: square; S_2: bottom

∴ find bottom, have what we need

$$\text{curl}\vec{F} = \begin{vmatrix} \uparrow & \updownarrow & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xyz & xy & x^2yz \end{vmatrix}$$

S_2: $z=-1$ w/ $-1 \leq x \leq 1$, $-1 \leq y \leq 1$

$= \langle x^2 z, -2xyz + xy, y - xz \rangle$

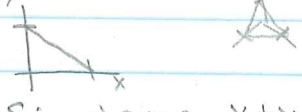
$\int_{-1}^{1}\int_{-1}^{1} (-x^2 z(0) - (xy - 2xyz)(0) + y - xz)\,dy\,dx$

$\int_{-1}^{1}\int_{-1}^{1} (y + x)\,dy\,dx = \int_{-1}^{1} \left(\frac{1}{2}y^2 + xy\right)\Big|_{-1}^{1} dx$

$= \int_{-1}^{1} \left(\frac{1}{2} + x - \frac{1}{2} + x\right) dx = x^2\Big|_{-1}^{1} = 1 - 1 = 0$

$\int_{-1}^{1}\int_{-1}^{1} (1)\,dy\,dx = 2(\text{area}) = 4$

7. $\vec{F} = \langle x+y^2, y+z^2, z+x^2 \rangle$





S: plane $x+y+z=1$

$z = 1-x-y$

$$\text{curl}\vec{F} = \begin{vmatrix} \uparrow & \updownarrow & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+y^2 & y+z^2 & z+x^2 \end{vmatrix}$$

$= \langle -2z, -2x, -2y \rangle$

$\iint_S (-(-2z)(-1) - (-2x)(-1) + (-2y))\,dy\,dx$

$= \int_0^1 \int_0^{1-x} (-2(1-x-y) - 2x - 2y)\,dy\,dx =$

9. $\vec{F} = \langle 2z, 4x, 5y \rangle$ C: $z = x+4$ and $x^2+y^2=4$

S: $z = x+4$ inside $x^2+y^2=4$, $r=2$

$$\text{curl}\,\vec{F} = \begin{vmatrix} \uparrow & \uparrow & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2z & 4x & 5y \end{vmatrix} = \langle 5, 2, 4 \rangle$$

$$\iint_S (-5(1) - 2(0) + 4)\, dy\, dx$$

$$-\iint_S 1\, dA = -1(\text{area}) = -(\pi \cdot 2^2) = -4\pi$$

(proj xy is circle)

11a. $\vec{F} = \langle x^2 z, xy^2, z^2 \rangle$ C: $x+y+z=1$, $x^2+y^2=9$

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl}\,\vec{F} \cdot d\vec{S}$$

S: $z = 1-x-y$ inside $x^2+y^2=9$

proj xy: circle $r=3$

$$\text{curl}\,\vec{F} = \begin{vmatrix} \uparrow & \uparrow & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 z & xy^2 & z^2 \end{vmatrix} = \langle 0, x^2, y^2 \rangle$$

$$\iint_S (0 - x^2(-1) + 0)\, dA$$

$$= \iint_S (x^2 + y^2)\, dA$$

$$= \int_0^{2\pi} d\theta \int_0^3 r\, dr \cdot r^2$$

$$= \frac{1}{4} \cdot 3^4 \cdot 2\pi = \frac{81\pi}{2}$$

13. $\vec{F} = \langle 3y, 4z, -6x \rangle$ S: $z = 9 - x^2 - y^2$, $z > 0$

C: circle $x^2+y^2=9$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (-27\sin^2 t + 0 + 0)\, dt$$

$\vec{r} = \langle 3\cos t, 3\sin t, 0 \rangle$

$\vec{r}' = \langle -3\sin t, 3\cos t, 0 \rangle$

$\vec{F}(\vec{r}) = \langle 9\sin t, 0, -18\cos t \rangle$

$$-27\int_0^{2\pi} \sin^2 t\, dt = -27\pi$$

$$\text{curl}\,\vec{F} = \begin{vmatrix} \uparrow & \uparrow & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3y & 4z & -6x \end{vmatrix} = \langle -4, 6, -3 \rangle$$

$$\iint_S (4(-2x) - 6(-2y) - 3)\, dy\, dx$$

$$\int_0^{2\pi} d\theta \int_0^3 r\, dr \cdot (-8r\cos\theta + 12 r\sin\theta - 3)$$

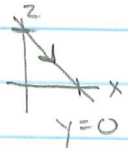
$$= -27\pi$$

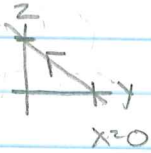


15. $\vec{F} = \langle y, z, x \rangle$ S: $z = 1 - x - y$, 1st oct, up

C: 3 segments

$\int_C \vec{F} \cdot d\vec{r}$ C_1: 

C_2: 

C_3: 

$\vec{r} = \langle t, 1-t, 0 \rangle$ $t: 1 \to 0$ | $\vec{r} = \langle t, 0, 1-t \rangle$ $t: 0 \to 1$ | $\vec{r} = \langle 0, t, 1-t \rangle$ $t: 1 \to 0$

$\vec{r}' = \langle 1, -1, 0 \rangle$ | $\vec{r}' = \langle 1, 0, -1 \rangle$ | $\vec{r}' = \langle 0, 1, -1 \rangle$

$\vec{F}(r) = \langle 1-t, 0, t \rangle$ | $\vec{F}(r) = \langle 0, 1-t, t \rangle$ | $\vec{F}(r) = \langle t, 1-t, 0 \rangle$

$\int_1^0 (1-t)\,dt$ | $\int_0^1 (-t)\,dt$ | $\int_1^0 (1-t)\,dt$

$-(1-\frac{1}{2})$ | $-\frac{1}{2}$ | $-(1-\frac{1}{2})$

$-\frac{1}{2}$ | | $-\frac{1}{2}$

$-3/2$

$\iint_S \operatorname{curl} \vec{F} \cdot d\vec{S} = \iint_S (1 \cdot (-1) + 1(-1) - 1)\,dy\,dx$ 

$\operatorname{curl} \vec{F} = \begin{vmatrix} \uparrow & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix}$

$= \langle -1, -1, -1 \rangle$

$= -\int_0^1 \int_0^{1-x} (1 + 1 + 1)\,dy\,dx$

$= -\int_0^1 \int_0^{1-x} 3\,dy\,dx$

$= -3\left(\frac{1}{2}\right)(1)(1) = \frac{-3}{2}$

→

18. $\oint_C (y+\sin x)\,dx + (z^2+\cos y)\,dy + x^3\,dz$

$x^2+y^2=1$ if $z=0$

$\vec{F} = \langle y+\sin x,\ z^2+\cos y,\ x^3\rangle$ C: $\vec{r} = \langle \sin t, \cos t, \sin 2t\rangle$

$0 \le t \le 2\pi$

$\operatorname{curl}\vec{F} = \langle -2z, -3x^2, -1\rangle$ S: $z = 2xy$

$\sin 2t = 2\sin t\cos t$

$\therefore z = 2xy$

$$-\iint_S \left(2z(2y) + 3x^2(2x) - 1\right) dy\,dx$$

$$-\iint_S \left(4y(2xy) + 6x^3 - 1\right) dy\,dx$$

$$-\iint_S \left(8xy^2 + 6x^3 - 1\right) dy\,dx$$

C clockwise $\therefore$ negative

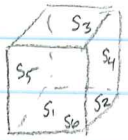
$$-\int_0^{2\pi} d\theta \int_0^1 r\,dr \cdot \left(8r^3\cos\theta\sin^2\theta + 6r^3\cos^3\theta - 1\right)$$

$$= \pi$$

HW

17.9 p.1150 #3,5,7*,11,13,19,22

3. $\vec{F} = \langle 3x, xy, 2xz \rangle$ cube: $x=0, x=1, y=0, y=1, z=0, z=1$



$\operatorname{div} \vec{F} = 3 + x + 2x = 3 + 3x$

$$\int_0^1 dz \int_0^1 dy \int_0^1 (3+3x)\,dx = (3x + \tfrac{3}{2}x^2)\Big|_0^1 = 3 + \tfrac{3}{2} = \boxed{\tfrac{9}{2}}$$

S_1: $\vec{n} = \uparrow$, $\vec{r} = \langle 1, y, z \rangle$, $\vec{F}(r) = \langle 3, y, 2z \rangle$

$\vec{F} \cdot \vec{n} = 3 + 0 + 0 = 3$

$$\iint_{S_1} 3\,dS = 3$$

S_2: $\vec{n} = \uparrow$, $\vec{r} = \langle x, 1, z \rangle$, $\vec{F}(r) = \langle 3x, x, 2xz \rangle$

$\vec{F} \cdot \vec{n} = 0 + x + 0 = x$

$$\iint_{S_2} x\,dS = \int_0^1 dx \int_0^1 x\,dz = 1/2$$

S_3: $\vec{n} = \hat{k}$, $\vec{r} = \langle x, y, 1 \rangle$, $\vec{F}(r) = \langle 3x, xy, 2x \rangle$

$\vec{F} \cdot \vec{n} = 2x$

$$\iint_{S_3} 2x\,dS = \int_0^1 dy \int_0^1 2x\,dx = 1$$

S_4: $\vec{n} = -\uparrow$, $\vec{r} = \langle 0, y, z \rangle$, $\vec{F}(r) = \langle 0, 0, 0 \rangle$

$\vec{F} \cdot \vec{n} = 0$

$$\iint_{S_4} 0\,dS = 0$$

S_5: $\vec{n} = -\uparrow$, $\vec{r} = \langle x, 0, z \rangle$, $\vec{F}(r) = \langle 3x, 0, 2xz \rangle$

$\vec{F} \cdot \vec{n} = 0$

$$\iint_{S_5} 0\,dS = 0$$

S_6: $\vec{n} = -\hat{k}$, $\vec{r} = \langle x, y, 0 \rangle$, $\vec{F}(r) = \langle 3x, xy, 0 \rangle$

$\vec{F} \cdot \vec{n} = 0$

$$\iint_{S_6} 0\,dS = 0$$

So $\iint_S \vec{F} \cdot d\vec{S} = 3 + \frac{1}{2} + 1 + 0 + 0 + 0 = \boxed{\frac{9}{2}}$



5. $\vec{F} = \langle xy, yz, zx \rangle$ cylinder: $x^2+y^2 \le 1, 0 \le z \le 1$

$\operatorname{div}\vec{F} = y+z+x$ $\int_0^1 dz \int_0^1 r\,dr \int_0^{2\pi} d\theta \cdot (r\sin\theta + z + r\cos\theta)$

$= \int_0^{2\pi}\int_0^1 r\left(r\sin\theta + \frac{1}{2} + r\cos\theta\right)dr$

$= \int_0^{2\pi} \left(\frac{1}{3}\sin\theta + \frac{1}{4} + \frac{1}{3}\cos\theta\right)d\theta$

$= \left(-\frac{1}{3}\cos\theta + \frac{1}{4}\theta + \frac{1}{3}\sin\theta\right)\Big|_0^{2\pi} = -\frac{1}{3} + \frac{\pi}{2} + \frac{1}{3} = \frac{\pi}{2}$

S_1: $\vec{n} = \hat{k}$, $\vec{r} = \langle x, y, 1\rangle$, $\vec{F}(r) = \langle xy, y, x\rangle$

$\vec{F}\cdot\vec{n} = x$

$\int_0^1 r\,dr \int_0^{2\pi} r\cos\theta\,d\theta = 0$

S_2: $\vec{r} = \langle \cos\theta, \sin\theta, z\rangle$, $\vec{r}_\theta = \langle -\sin\theta, \cos\theta, 0\rangle$

$\vec{r}_z = \langle 0, 0, 1\rangle$

$\vec{r}_\theta \times \vec{r}_z = \langle \cos\theta, \sin\theta, 0\rangle$ $\vec{F}(r) = \langle \cos\theta\sin\theta, z\sin\theta, z\cos\theta\rangle$

$\int_0^{2\pi}\int_0^1 \langle \cos^2\theta\sin\theta + z\sin^2\theta + 0\rangle\, dz\, d\theta$

$= \int_0^{2\pi} \left(\cos^2\theta\sin\theta + \frac{1}{2}\sin^2\theta\right)d\theta = \left(-\frac{1}{3}\cos^3\theta + \frac{1}{4}\theta - \frac{1}{8}\sin 2\theta\right)\Big|_0^{2\pi}$

$u = \cos\theta$ $\quad \frac{1}{4}(1-\cos 2\theta)$

$= \pi/2$

S_3: $\vec{n} = -\hat{k}$, $\vec{r} = \langle x, y, 0\rangle$, $\vec{F}(r) = \langle xy, 0, 0\rangle$

$\vec{F}\cdot\vec{n} = 0$

$\iint_{S_3} 0\,dS = 0$

$\iint_S \vec{F}\cdot d\vec{S} = 0 + \frac{\pi}{2} + 0 = \frac{\pi}{2}$

7. $\vec{F} = \langle 3y^2z^3, 9x^2yz^2, -4xy^2\rangle$ $\operatorname{div}\vec{F} = 0 + 9x^2z^2 + 0 = 9x^2z^2$

$\int_{-1}^1\int_{-1}^1\int_{-1}^1 9x^2z^2\,dV = 4\int_0^1\int_0^1\int_{-1}^1 9x^2z^2\,dx\,dy\,dz = 4\int_{-1}^1 3x^2 = 4x^3\Big|_{-1}^1$

$= 4(1+1) = 8$

11. $\vec{F} = \langle 3xy^2, xe^z, z^3 \rangle$ $\quad y^2+z^2=1, \quad -1 \le x \le 2$

$\operatorname{div} \vec{F} = 3y^2 + 0 + 3z^2 = 3(y^2+z^2)$

$$\int_{-1}^{2} dx \int_0^1 r\,dr \cdot \int_0^{2\pi} 3(r^2\sin^2\theta + r^2\cos^2\theta)\,d\theta$$

$$= 3 \cdot 3 \cdot 2\pi \cdot \int_0^1 r^3 = 18\pi \cdot \frac{1}{4} = \frac{9\pi}{2}$$

13. $\vec{F} = \langle x^3, y^3, z^3 \rangle$, sphere $\rho = 1$ $\quad (x^2+y^2+z^2=1)$

$\operatorname{div} \vec{F} = 3x^2 + 3y^2 + 3z^2$

$$3\int_0^1 \rho^4\,d\rho \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi\,d\phi = 3 \cdot \frac{1}{5} \cdot 2\pi \cdot (-\cos\theta)\Big|_0^{\pi}$$

$$= \frac{6\pi}{5}(1+1) = 12\pi/5$$

19. $\vec{F} = \langle z^2x, \frac{1}{3}y^3 + \tan z, x^2z + y^2 \rangle$, top half $x^2+y^2+z^2=1$

S_1: disc $\quad \vec{n} = -\hat{k}$, $\quad \vec{r} = \langle x, y, 0 \rangle$, $\vec{F}(r) = \langle 0, \frac{1}{3}y^3, y^2 \rangle$

$\vec{F} \cdot \vec{n} = -y^2$

$$\iint_{S_1} -y^2\,dS = -\int_0^1 r\,dr \int_0^{2\pi} r^2\sin^2\theta\,d\theta$$

$$= -\frac{1}{4}\int_0^{2\pi} \left(1 - \frac{\cos 2\theta}{2}\right) d\theta$$

$$= -\frac{1}{4}\left(\frac{1}{2}\theta - \frac{1}{2}\sin\theta\right)\Big|_0^{2\pi} = -\frac{1}{4}(\pi) = -\frac{\pi}{4}$$

S_2: hemisphere

$\operatorname{div} \vec{F} = z^2 + y^2 + x^2$ $\quad \int_0^{2\pi} d\theta \int_0^{\pi/2} \sin\phi \int_0^1 \rho^4\,d\rho = 2\pi(-\cos\phi)\Big|_0^{\pi/2} \cdot \frac{1}{5}$

$$= \frac{2\pi}{5}(0+1) = 2\pi/5$$

$$\iint_{S_2} F \cdot dS = \iint_{S} F \cdot dS + \iint_{S_1} F \cdot dS$$

$$\iint_{S} F \cdot dS = \frac{2\pi}{5} + \frac{\pi}{4} = \frac{13\pi}{20}$$

22. $\iint_S (2x+2y+z^2)\,dS$ $\quad S: x^2+y^2+z^2=1$

$\vec{F}\cdot\vec{n} = 2x+2y+z^2$, $\vec{F}$?

sphere: $\vec{n} = \langle 2x, 2y, 2z\rangle$ and unit $\vec{n} = \dfrac{2\langle x,y,z\rangle}{\sqrt{4x^2+4y^2+4z^2}}$

$= \langle x, y, z\rangle$

$\vec{F}\cdot\langle x,y,z\rangle = \langle 2x+2y+z^2\rangle$

so $\vec{F} = \langle 2, 2, z\rangle$

$$\iint_S F\,dS = \iiint_E \nabla\cdot F\,dV = \iiint 1\,dV = \frac{4\pi}{3}$$